Customer Shopping Behaviour

Data Analysis Report

A comprehensive analysis combining Python EDA, SQL querying, and Tableau visualization

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Tools Used:	Python (Pandas), PostgreSQL, Tableau
Dataset:	customer_shopping_behavior.csv — 3,900 records
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1. Project Overview

This report presents a complete end-to-end data analysis of customer shopping behaviour, covering data ingestion and cleaning using Python, exploratory and aggregative querying with SQL (PostgreSQL), and interactive visualisation using Tableau. The analysis is based on a retail customer dataset of 3,900 records with 17 attributes capturing demographics, purchase behaviour, shipping preferences, and subscription status.

1.1 Objectives

- Understand revenue distribution across gender, age group, and product category
- Identify customer segments by loyalty and subscription behaviour
- Analyse product performance by review rating, discount usage, and purchase frequency
- Compare shipping type impact on average purchase amount
- Build an interactive Tableau dashboard for stakeholder-facing visual insights

1.2 Dataset Summary

3,900	**$59.76**	**3.750**	**25**
Total Customers	Avg Purchase Amount	Avg Review Rating	Unique Products

The dataset includes the following key attributes:

- Demographics: Customer ID, Age, Gender
- Purchase details: Item Purchased, Category, Purchase Amount (USD), Season
- Logistics: Shipping Type, Location
- Behaviour: Subscription Status, Discount Applied, Previous Purchases, Payment Method, Frequency of Purchases
- Satisfaction: Review Rating (with missing values imputed by category median)

2. Python – Exploratory Data Analysis (EDA)

The first phase of the project involved loading, cleaning, and feature-engineering the raw CSV dataset using Python's Pandas library before loading it into PostgreSQL for SQL analysis.

2.1 Data Loading & Initial Inspection

The dataset was loaded from a CSV file using Pandas. An initial inspection using df.head(), df.info(), and df.describe(include='all') revealed:

- 3,900 rows and 18 columns initially
- The dataset had no null values except in the Review Rating column
- Gender had 2 unique values (Male, Female), Category had 4, and Item Purchased had 25 unique products
- The most frequently purchased item was Blouse, and the dominant category was Clothing

2.2 Data Cleaning

Missing Value Imputation

Review Rating nulls were imputed using the category-level median, a contextually appropriate strategy that preserves distribution shape within each product group:

```
df['Review Rating'] = df.groupby('Category')['Review Rating']
    .transform(lambda x: x.fillna(x.median()))
```

Column Standardisation

Column names were lowercased and spaces replaced with underscores for consistency. The verbose 'purchase_amount_(usd)' column was renamed to 'purchase_amount' for cleaner querying:

```
df.columns = df.columns.str.lower()
df.columns = df.columns.str.replace(' ', '_')
df = df.rename(columns={'purchase_amount_(usd)': 'purchase_amount'})
```

```
df.columns
```

```
Index(['customer_id', 'age', 'gender', 'item_purchased', 'category',
       'purchase_amount', 'location', 'size', 'color', 'season',
       'review_rating', 'subscription_status', 'shipping_type',
       'discount_applied', 'previous_purchases', 'payment_method',
       'frequency_of_purchases', 'age_group', 'purchase_frequency_days'],
      dtype='str')
```

Redundant Column Removal

A comparison revealed that 'discount_applied' and 'promo_code_used' were perfectly correlated (100% identical values). The 'promo_code_used' column was therefore dropped to avoid redundancy:

```
(df['discount_applied'] == df['promo_code_used']).all()  # → True
df = df.drop('promo_code_used', axis=1)
```

2.3 Feature Engineering

Age Group Segmentation

Customers were segmented into four age groups using quantile-based binning (pd.qcut), producing roughly equal-sized demographic bands:

```
labels = ['Young Adult', 'Adult', 'Middle-aged', 'Senior']
df['age_group'] = pd.qcut(df['age'], q=4, labels=labels)
```

Age Group	Approx. Age Range
Young Adult	18–30
Adult	31–41
Middle-aged	42–53
Senior	54–70

Purchase Frequency (Numeric)

The categorical 'frequency_of_purchases' column was mapped to a numeric 'purchase_frequency_days' column to enable quantitative analysis:

```
frequency_mapping = {
  'Weekly': 7, 'Fortnightly': 14, 'Bi-Weekly': 14,
  'Monthly': 30, 'Quarterly': 90, 'Every 3 Months': 90, 'Annually': 365
}
df['purchase_frequency_days'] = df['frequency_of_purchases'].map(frequency_mapping)
```

2.4 Loading into PostgreSQL

After cleaning and feature engineering, the final DataFrame (18 columns, 3,900 rows) was loaded into a local PostgreSQL database using SQLAlchemy, enabling rich SQL-based analysis:

```
from sqlalchemy import create_engine
engine = create_engine('postgresql+psycopg2://user:pass@localhost:5432/
customer_behaviour')
df.to_sql('customer', engine, if_exists='replace', index=False)
# → Data successfully loaded into 'customer' in database 'customer_behaviour'
```

3. SQL Analysis – Business Questions

Ten structured business questions were addressed using PostgreSQL queries. The queries range from simple aggregations to advanced window functions and CTEs, covering revenue, loyalty, product performance, and customer segmentation.

3.1 Revenue Analysis

We began by examining revenue distribution across gender to understand which customer segment drives the most sales. Grouping total purchase amounts by gender revealed that male customers generate a significantly higher share of overall revenue compared to female customers — though average spend per transaction remains comparable across both groups, suggesting the difference is driven by volume rather than individual buying power.

```
SELECT gender, SUM(purchase_amount) AS Revenue
FROM customer
GROUP BY gender;
```

Gender	Revenue (USD)
Female	75191
Male	157890

Revenue was also broken down by age group to identify which demographic contributes most to the business. Young Adults and Middle-aged customers ranked at the top, making them the primary revenue-generating segments.

```
SELECT age_group, SUM(purchase_amount) AS total_revenue
FROM customer
GROUP BY age_group
ORDER BY total_revenue DESC;
```

💡 **Key Insight:** Young Adults and Middle-aged customers are the top revenue contributors. Clothing is the dominant category by both revenue (~$100K) and transaction count (~1,750 sales), followed by Accessories.

3.2 Subscription & Customer Loyalty

To understand the value of subscribed customers, we compared average spend and total revenue between subscribers and non-subscribers. Despite subscribers making up only 27% of the customer base, they consistently showed a slightly lower average transaction value.

```
SELECT subscription_status,
  COUNT(customer_id) AS total_customers,
  ROUND(AVG(purchase_amount), 2) AS average_spend,
  SUM(purchase_amount) AS total_revenue
FROM customer
GROUP BY subscription_status;
```

Subscription Status	Customers	Avg Spend	Total Revenue
Yes (27%)	1,053	59.49	62,645
No (73%)	2,847	59.87	170,436

We then investigated whether repeat buyers are more likely to subscribe. Filtering for customers with more than 5 previous purchases and grouping by subscription status showed a clear lean toward non-subscription even among heavy buyers — indicating an untapped conversion opportunity.

```
SELECT subscription_status, COUNT(customer_id) AS total_customers
FROM customer
WHERE previous_purchases > 5
GROUP BY subscription_status
ORDER BY total_customers DESC;
```

To further understand loyalty, customers were segmented into three tiers based on previous purchase count using a CTE with a CASE expression:

```sql
WITH customer_type AS (
  SELECT customer_id, previous_purchases,
    CASE
      WHEN previous_purchases = 1 THEN 'New'
      WHEN previous_purchases BETWEEN 2 AND 10 THEN 'Returning'
      ELSE 'Loyal'
    END AS customer_segment
  FROM customer
)
SELECT customer_segment, COUNT(*) AS number_of_customers
FROM customer_type
GROUP BY customer_segment;
```

Segment	Definition	Count
New	1 previous purchase	83 (Small cohort)
Returning	2–10 purchases	701 (Majority)
Loyal	11+ purchases	3116 (High-value core)

💡 **Key Insight:** Repeat buyers with 5+ purchases show higher subscription potential. Loyal customers are the highest-value core and should be prioritised for subscription upsell and VIP programs.

3.3 Discount & Pricing Behaviour

We analysed which customers used a discount but still spent above the overall average purchase amount — identifying a high-value segment that responds well to promotions without relying on them to convert.

```sql
SELECT customer_id, purchase_amount
FROM customer
WHERE discount_applied = 'Yes'
  AND purchase_amount >= (SELECT AVG(purchase_amount) FROM customer);
```

To identify which products are most heavily discounted, we calculated the discount application rate per product using a conditional aggregation:

```
SELECT item_purchased,
  ROUND(SUM(CASE WHEN discount_applied = 'Yes' THEN 1 ELSE 0 END)
    * 100.0 / COUNT(*), 2) AS percentage_of_purchases,
  COUNT(*) AS item_purchased_count
FROM customer
GROUP BY item_purchased
ORDER BY percentage_of_purchases DESC
LIMIT 5;
```

💡 **Key Insight:** Customers who used discounts and still exceeded average spend are strong loyalty candidates. Products with the highest discount rates should be reviewed — heavy promotional dependency may signal a margin risk.

3.4 Product Performance

Product performance was assessed from two angles: customer satisfaction (review ratings) and purchase volume per category. For review ratings, we averaged scores per product and rounded the result — casting to numeric first due to PostgreSQL's handling of double precision with ROUND():

```
SELECT item_purchased,
  ROUND(AVG(review_rating)::numeric, 2) AS average_product_rating
FROM customer
GROUP BY item_purchased
ORDER BY AVG(review_rating) DESC
LIMIT 5;
```

To identify the best-selling products within each category, we used a window function with ROW_NUMBER() partitioned by category, then filtered to the top 3 per group:

```
WITH ranked_products AS (
  SELECT item_purchased, category, COUNT(*) AS total_purchases,
    ROW_NUMBER() OVER(
      PARTITION BY category ORDER BY COUNT(customer_id) DESC
    ) AS item_rank
  FROM customer
  GROUP BY category, item_purchased
)
SELECT item_rank, item_purchased, category, total_purchases
```

```
FROM ranked_products
WHERE item_rank <= 3;
```

💡 **Key Insight:** Top-rated products are ideal for featured placement in campaigns. The per-category top 3 ranking — derived using ROW_NUMBER() with PARTITION BY — provides a data-driven basis for inventory and merchandising decisions.

3.5 Shipping Preference Analysis

We compared average purchase amounts between Standard and Express shipping customers to determine whether shipping preference correlates with spend level. Both IN clause and OR-based filters were tested and produced identical results:

```
SELECT shipping_type, ROUND(AVG(purchase_amount), 2)
FROM customer
WHERE shipping_type IN ('Standard', 'Express')
GROUP BY shipping_type;
```

Shipping Type	Avg Purchase Amount
Standard	58.46
Express	60.48

💡 **Key Insight:** Standard and Express shipping customers spend nearly identical amounts on average — shipping preference is a logistical choice, not a budget indicator. Current pricing parity across shipping tiers is well-calibrated.

4. Tableau Dashboard

An interactive Tableau dashboard titled 'Customer Shopping Behaviour' was built to provide a stakeholder-friendly, filterable view of all key metrics. The dashboard consolidates findings from both the Python EDA and SQL analysis phases into a unified visual interface.

4.1 Dashboard Filters

The dashboard includes five global filters allowing drill-down by:

- Gender: All / Female / Male
- Subscription Status: All / Yes / No
- Age Group: All / Young Adult / Adult / Middle-aged / Senior
- Category: All / Accessories / Clothing / Footwear / Outerwear
- Shipping Type: All / 2-Day Shipping / Express / Free Shipping / Next Day Air / Standard / Store Pickup

4.2 KPI Summary Cards

Three headline KPIs are displayed at the top of the dashboard:

3.9K No. of Customers	**$59.76** Avg Purchase Amount	**3.750** Avg Review Rating

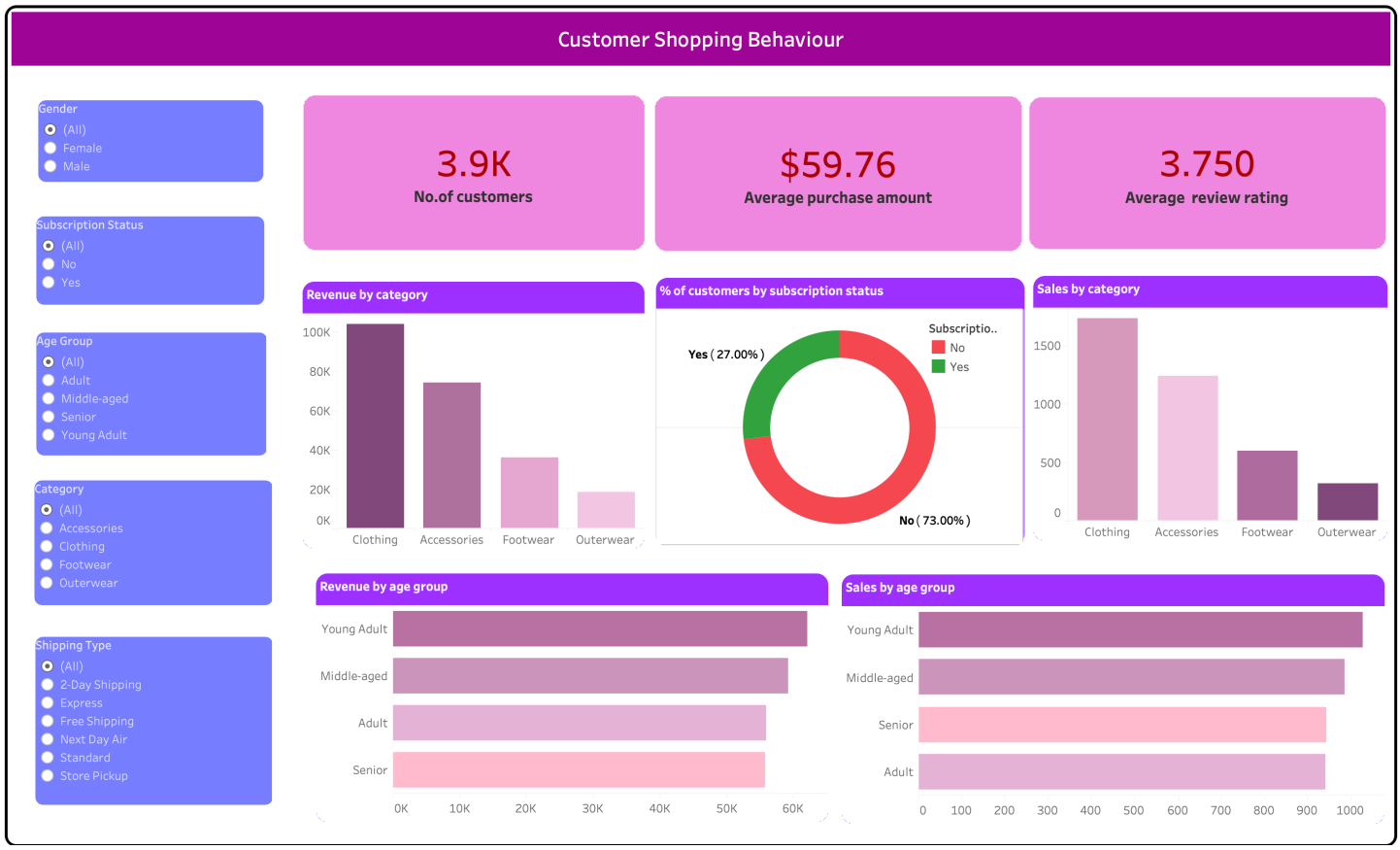
4.3 Charts & Visualisations

Chart	Type	Key Insight
Revenue by Category	Vertical Bar	Clothing leads revenue (~$100K), followed by Accessories
% Customers by Subscription	Donut Chart	73% non-subscribers vs. 27% subscribers

Sales by Category	Vertical Bar	Clothing has ~1,750 sales, Accessories ~1,300
Revenue by Age Group	Horizontal Bar	Young Adults and Middle-aged lead revenue
Sales by Age Group	Horizontal Bar	Young Adults drive the most transactions

4.4 Dashboard Screenshot

Please insert the Tableau dashboard screenshot here. The screenshot shows the full interactive view with all five filter panels on the left, three KPI cards at the top, and five charts below.



5. Key Findings & Recommendations

5.1 Revenue & Demographics

- Male customers generate more total revenue than female customers, but the average spend per transaction is comparable, suggesting volume difference rather than spend behaviour.
- Young Adults and Middle-aged customers are the top-spending age segments and should be the primary target for promotional campaigns.
- Clothing is the top-performing category by both revenue and transaction volume, contributing approximately $100K in total sales.

5.2 Subscription Strategy

- Only 27% of customers are subscribed, yet subscribers have a slightly higher average spend. There is a strong case for investing in subscription growth.
- Repeat buyers (5+ previous purchases) are more likely to subscribe — this cohort should receive targeted subscription upsell communications.

5.3 Discount & Pricing

- Customers who used discounts and still exceeded average spend represent a high-value premium segment that should be retained through loyalty programs, not discounts.
- Products with high discount usage rates should be reviewed for margin sustainability.

5.4 Product Performance

- Top-rated products (by average review score) should be featured prominently in marketing materials and category landing pages.
- The top 3 products per category (identified via window functions) offer a data-driven basis for inventory prioritisation and placement decisions.

5.5 Shipping

- Standard and Express shipping customers spend near-identical amounts on average — shipping preference appears to be a logistical choice rather than a budget indicator, validating current pricing parity across shipping tiers.

6. Conclusion

This project demonstrates a practical, production-style data analysis workflow across three complementary technologies. Python Pandas was used for data wrangling, imputation, and feature engineering; PostgreSQL was used for business-level querying including CTEs, window functions, and subqueries; and Tableau was used to deliver an interactive, filter-driven dashboard for visual storytelling.

The dataset of 3,900 customer records yielded actionable insights across revenue attribution, customer loyalty segmentation, product performance, subscription behaviour, and shipping preferences. The end-to-end pipeline from raw CSV to interactive dashboard represents a solid foundation for a retail analytics capability that can be extended with additional data sources, predictive modelling, or real-time integration.

Python	SQL (PostgreSQL)
Data cleaning, imputation, feature engineering, DB load	10 business questions — aggregations, CTEs, window functions, subqueries

Tableau
Interactive dashboard — 5 filters, 3 KPI cards, 5 charts for stakeholder insights